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Q1. Harshad number

ANSWER

```
n=int(input("Enter a number"))
def harshad(x):
    return x % sum(map(int,str(x)))==0
if harshad(n):
    print("Harshad Number")
else:
    print("Not a harshad number")
print("Harshad numbers(1-500):")
for i in range(1,501):
    if harshad(i):
        print(i,end=" ")
```

Q2. GCD

CODE

```
a=int(input("Enter first:\n"))
b=int(input("Enter second:\n"))
while b!=0:
    temp=b
    b=a%b
    a=temp
print("gcd is",a)
```

INPUT

```
24
36
```

OUTPUT

```
Enter first:
Enter second:
gcd is 12
```

Q3. sum to a target value

ANSWER

```
def find_pairs(1st,target):
    seen=set()
    pairs=[]
    for num in 1st:
        compliment=target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data=[int(x) for x in input("Enter numbers: ").split()]
t=int(input("Target:"))
result=find_pairs(data,t)
print("Pairs:",result)
```

Q4. Bitwise Operators

ANSWER

```
a=0b1101
b=0o25
c=0x1A
result=(a&b)|c
print(f"a={a},b={b},c={c},result={result}")
print(bin(reslt),oct(result),hex(result))
print(result>>2,result<<1,~result)
```

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<=arr[mid]:
            if target<=arr[low] and target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if target>arr[mid] and target<=arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=int(input("Target:\n"))
result=rotated_binary_search(data,t)
print("Found at",result)
```

INPUT

6

OUTPUT

Target:
Found at 2